TUTORIAL 2

IC Design Place and Route

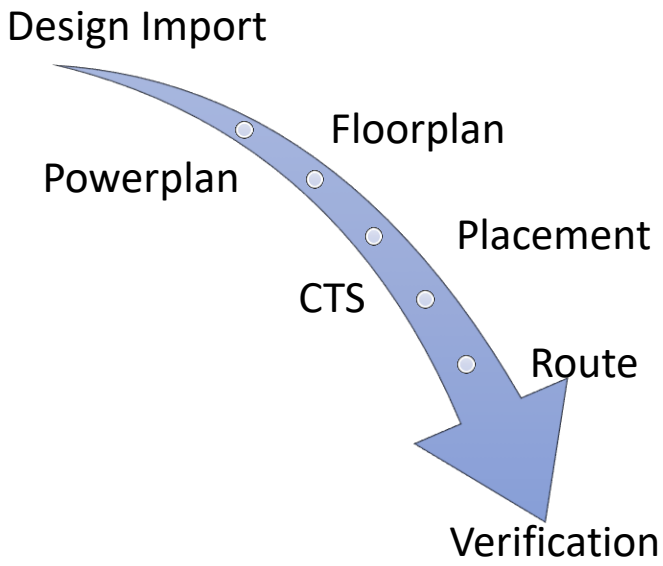
SUBMITTED BY

Kamrul Hasan Ovi
Matriculation Number: 6335212

1 Goals

The goal of this tutorial to learn the physical design in more details. In this tutorial, we developed for a small single-core RISC-V SoC called PULPino.

The design steps include

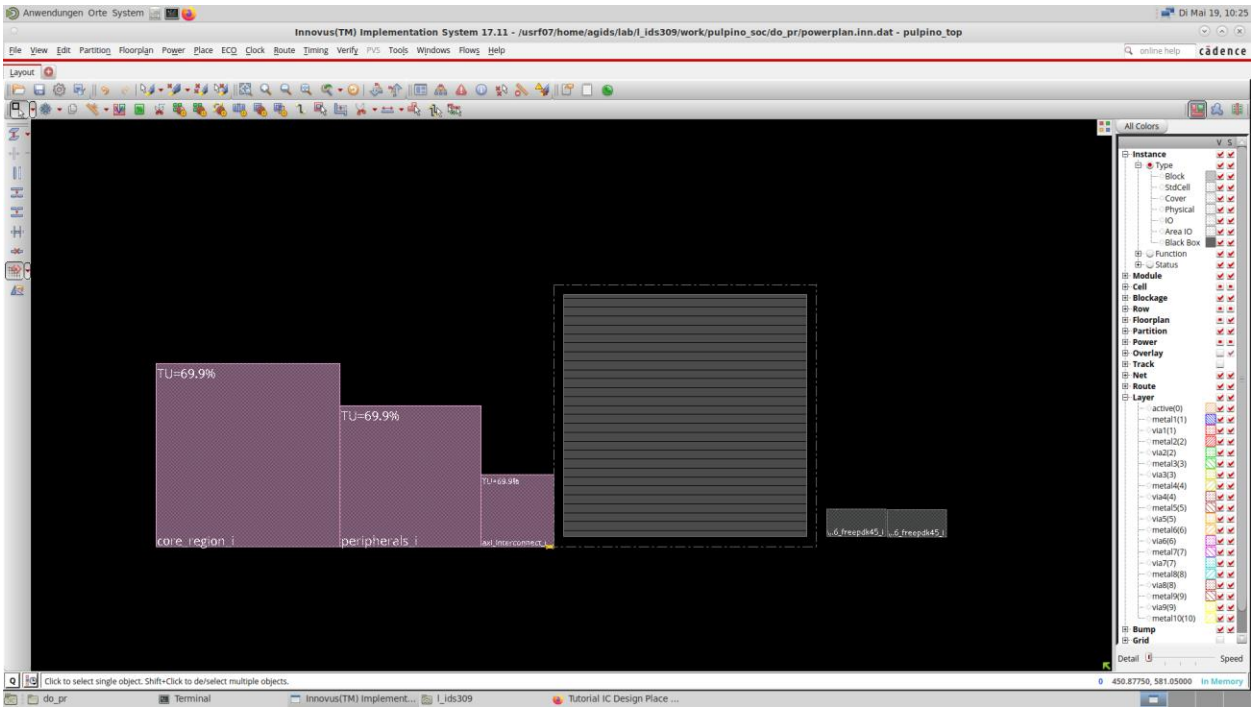


2 Design Process

2.1 Import Design

Inputs Required for import a design

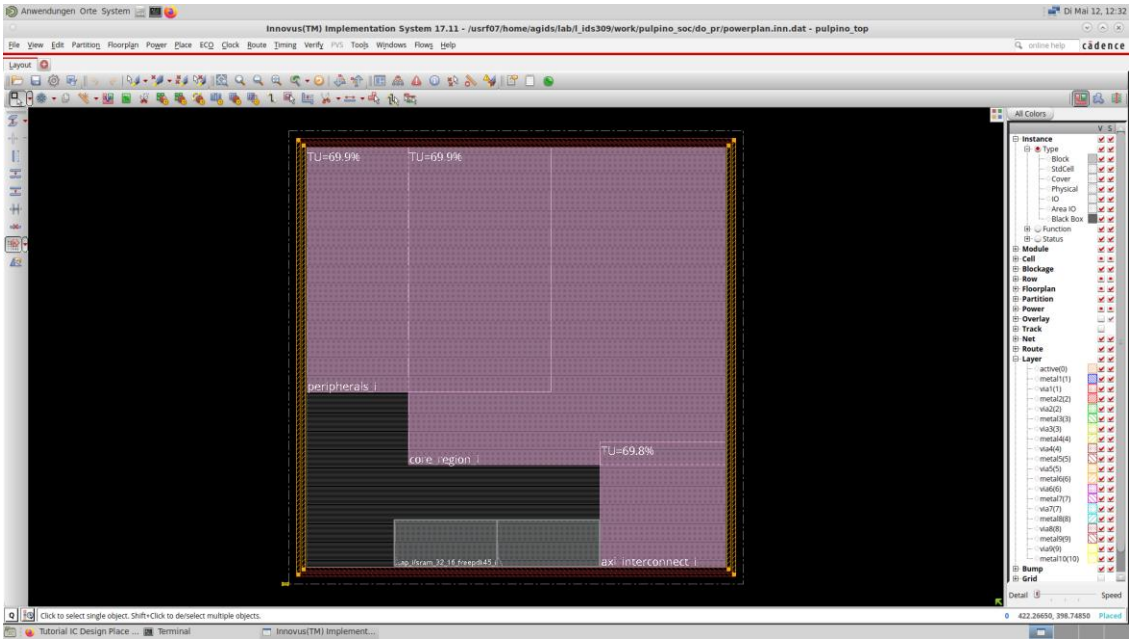
- **Netlist (.v or .vhd)** → It is the combination of sequential elements and their logical Connectivity
- **Synopsys Design Constraints (SDC)** →In the SDC file, Constraints related information are mentioned, related to design rule constraints, optimization constraints, timing constraints
- **Liberty Timing File (.lib)** → it contains timing related information, for example cell type and Functionality, Delay Models. pin/ Cell timings and design rules, PVT conditions, power details.
- **Library Exchange Format (LEF)** → LEF file contains physical information of the designs. LEF contains cell name, Shape, Size, orientation & class, Port/pin name, Direction and layout Geometrics, Obstruction/Blockages.
- **Multiple mode multiple corner file (MMMC.view)**



2.2 Floorplan

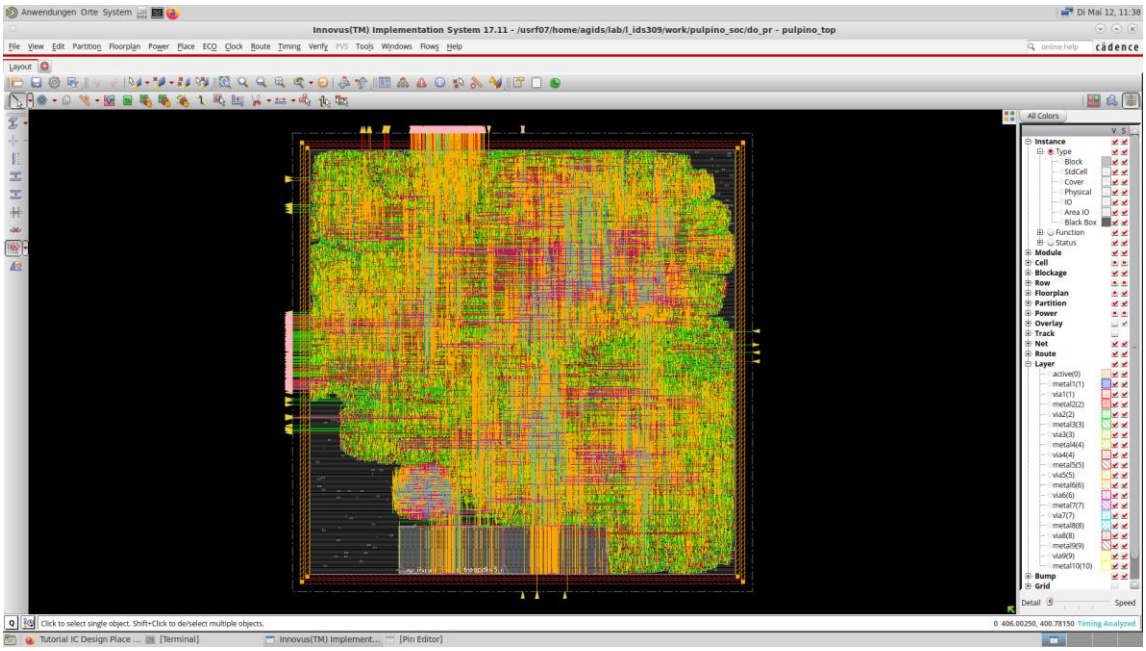
Steps in Floorplan

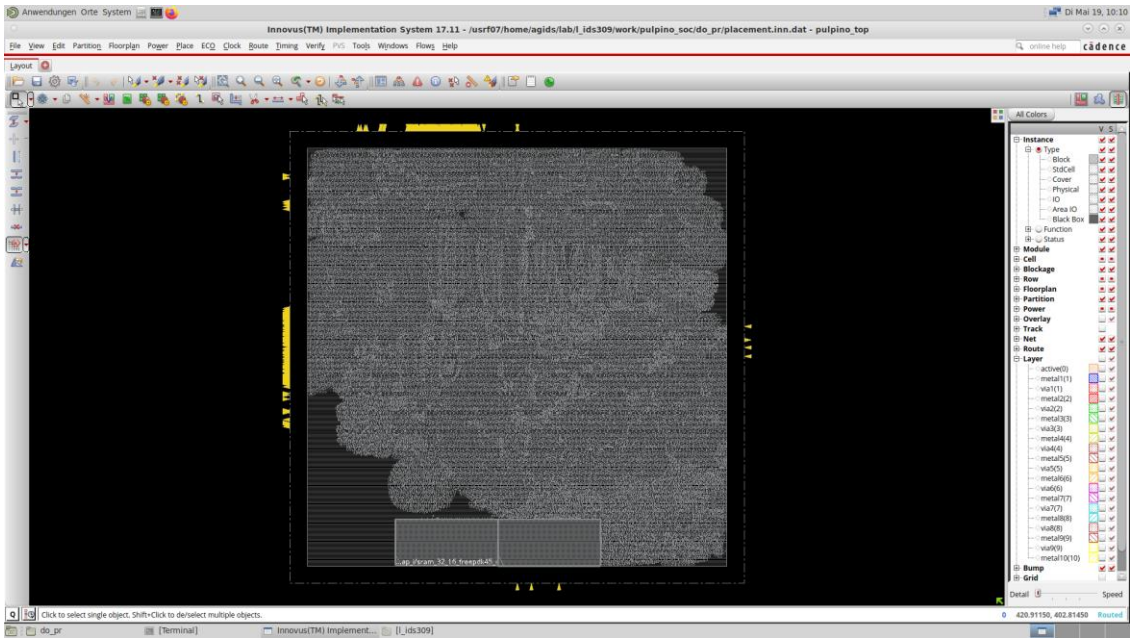
- Initialize with Chip & Core Aspect Ratio
- Initialize with Core Utilization
- Power Rings
- Pins /Pads Placement
- Macro Placement



2.3 Placement

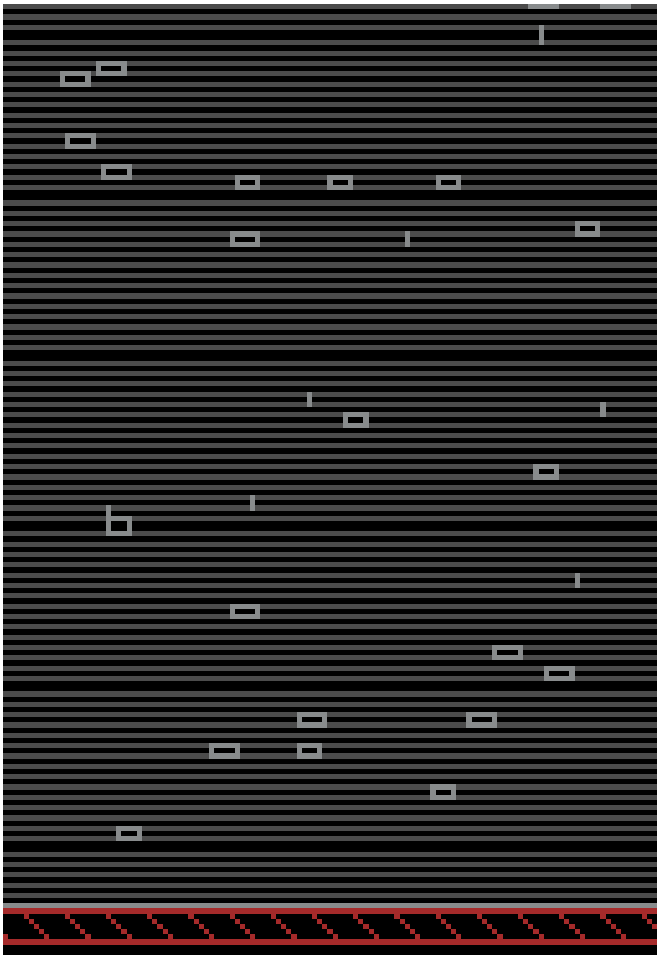
Placement is the process of placing standard cell in the design. The tools determine the location of each standard cell on the dies. the tool places these based on the algorithms which it used internally. Placement does not just place the standard cells available in the synthesized netlist. It also optimizes the design. Placement also determines the routability of design.





2.3.1 Spare Cell Placement

Spare cells are used to modify or improve the functionality of a chip. They are extra cells and placed at regular intervals in the chip. They are not timing-critical. When the design is tapped out and afterward there is a requirement to fix some bugs, then it is not possible to fix if we have not provided spare cells. This kind of floating cell is added in design so that after tape out if some bugs are needed to fix, we can make use of it.
There would be an insertion of approximately 5% spare cells in the whole design, not more than that.
Spare cells are connected to VDD/VSS when they are placed in the design and their output is left floating and if they are required to use their inputs are disconnected from VDD/VSS and connected to functional logic in Engineering Change order (ECO) mode.

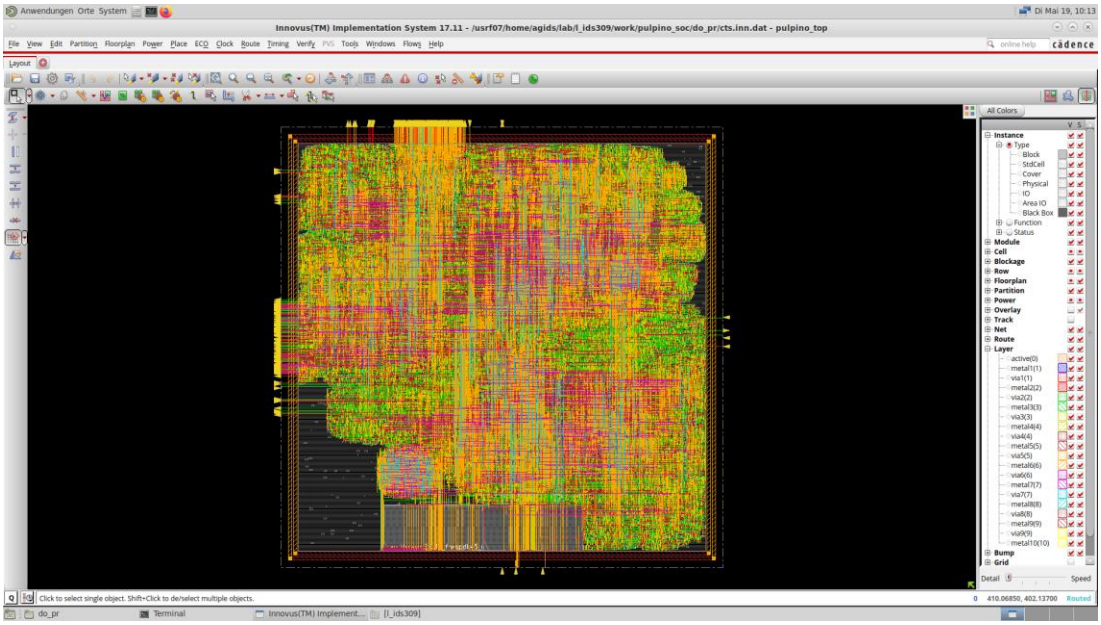


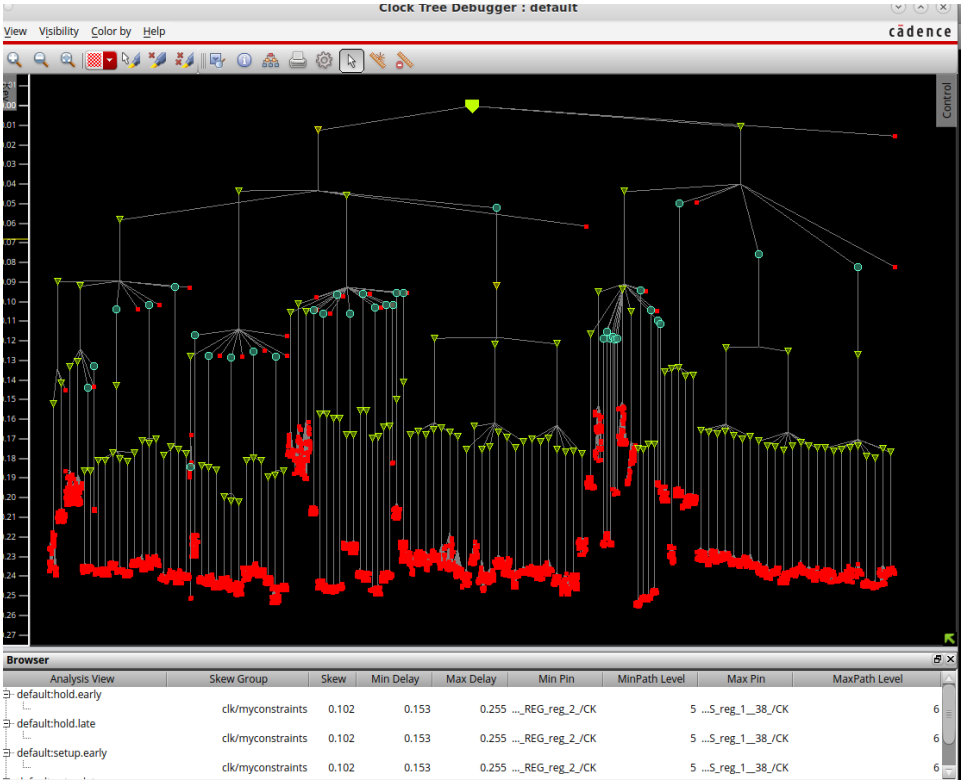
Placement Outputs

- Timing report
- Design with all std cells placed in core area

2.4 CTS

Clock Tree Synthesis is one of the most important stages in PnR. The process of distributing the clock and balancing the load is called CTS. CTS is the process of insertion of buffers or inverters along the clock paths if ASIC design in order to achieve zero/minimum slew or balanced skew.



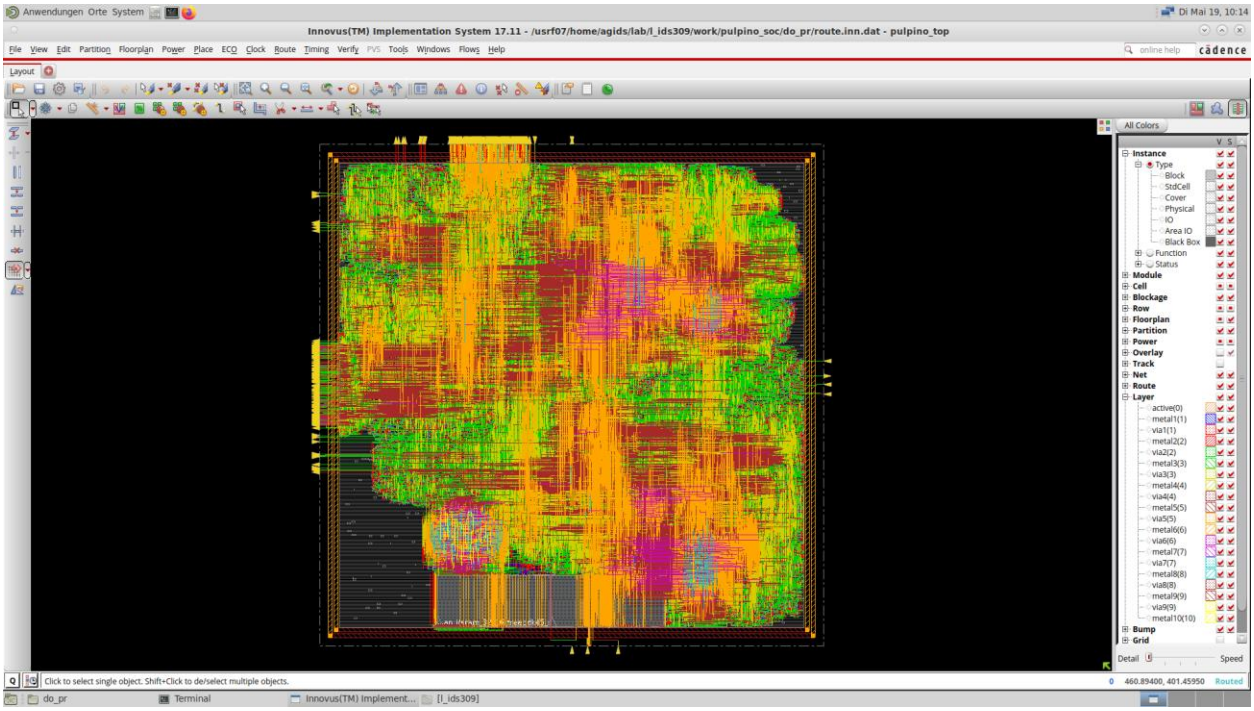


CTS output

- Timing report
- Skew report
- Congestion report

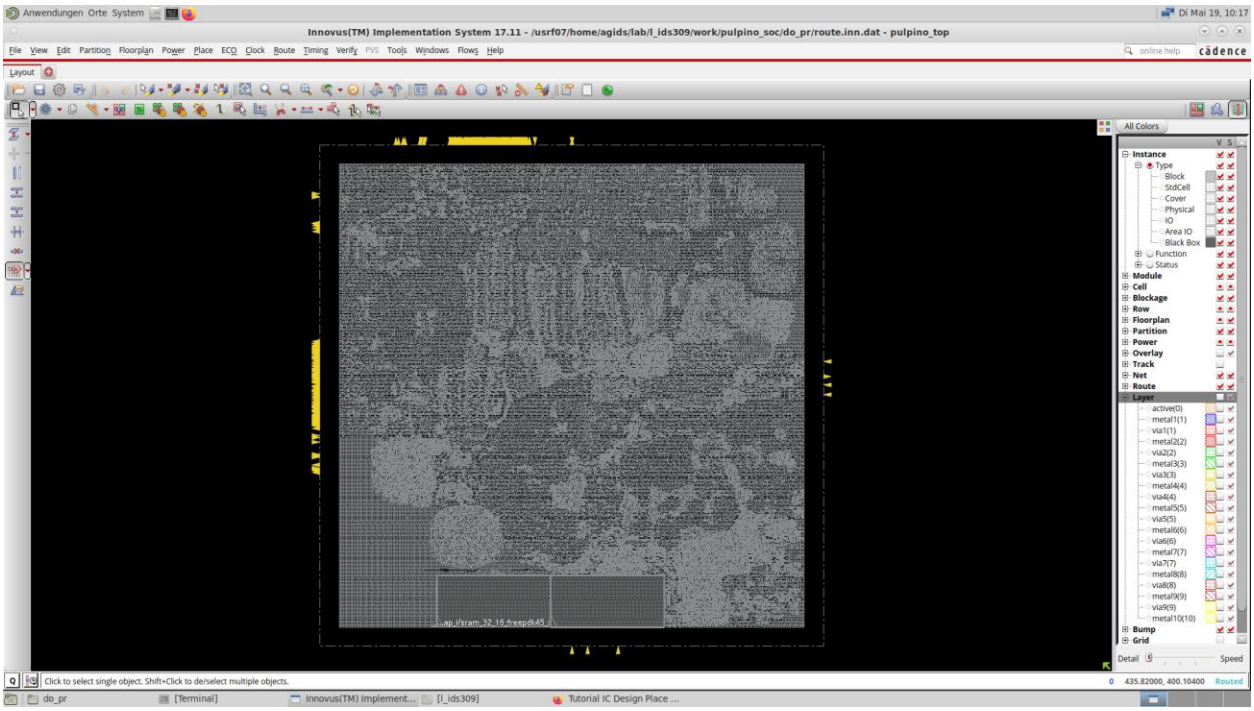
2.5 Routing

Making physical connections between signal pins and the standard cells ae called routing. Electrical connections using metal layers and vias are created in the layout, defined by the logical connections present in the netlist. After CTS, we have information of all the [laced cells and clock tree buffers/inverters and I/O pins, the tool relies I this information to electrically complete all connection defined in the netlist.



2.5.1 Filler Cells Placement

- Fill empty space in the standard-cell rows with filler cells.
- Filler cells have no logical connectivity. these cells are provided continuity in the rows for VDD and VSS nets and it also contains substrate nwell connection to improve substrate biasing.



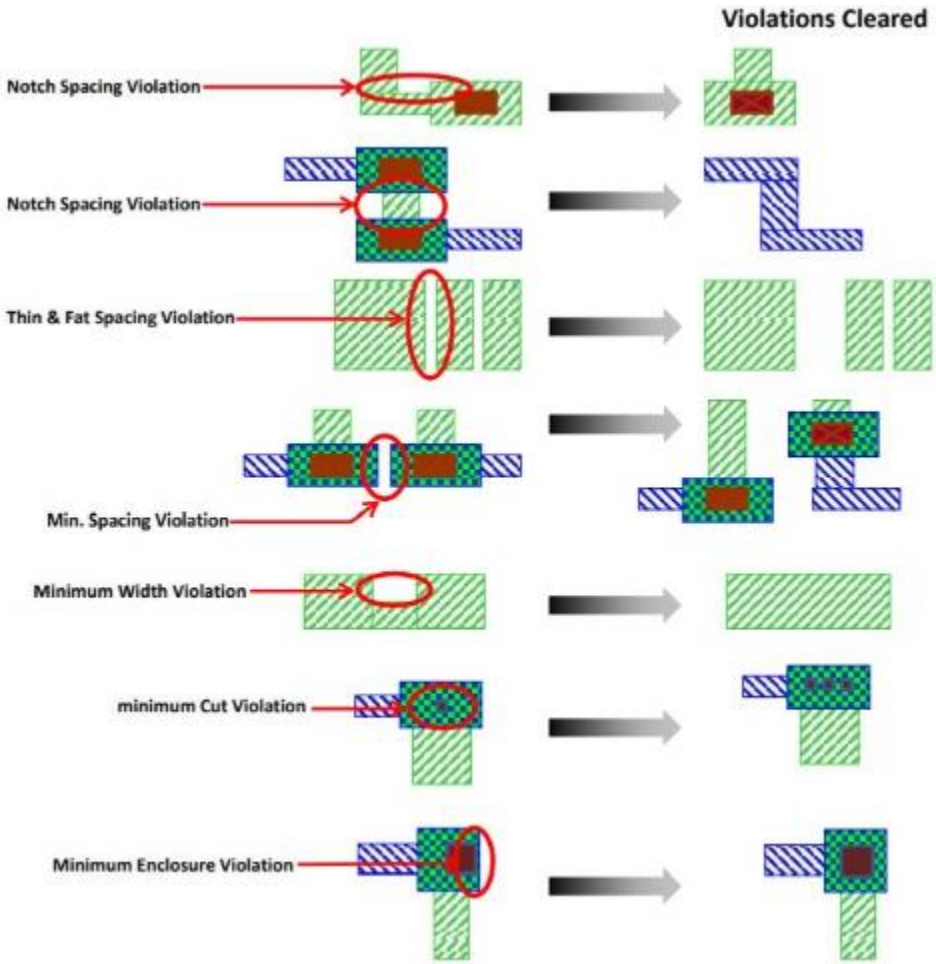
Outputs of Routing

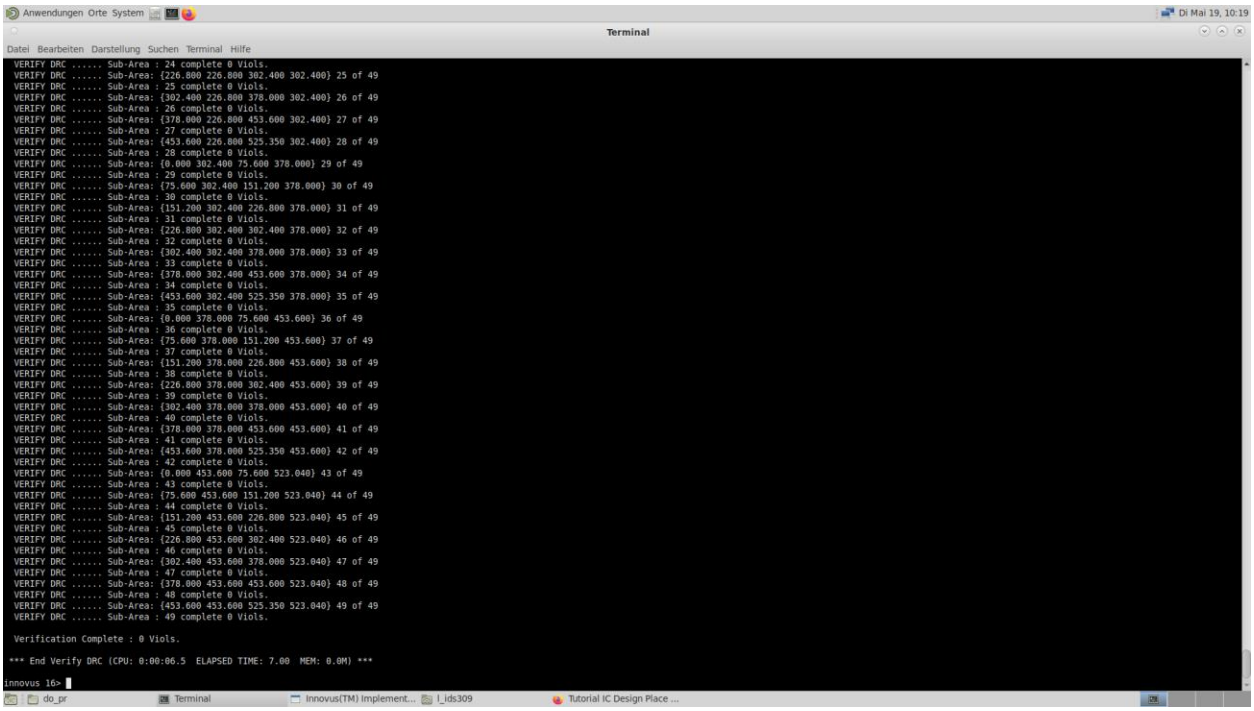
- Timing report
- Geometric layouts of all nets

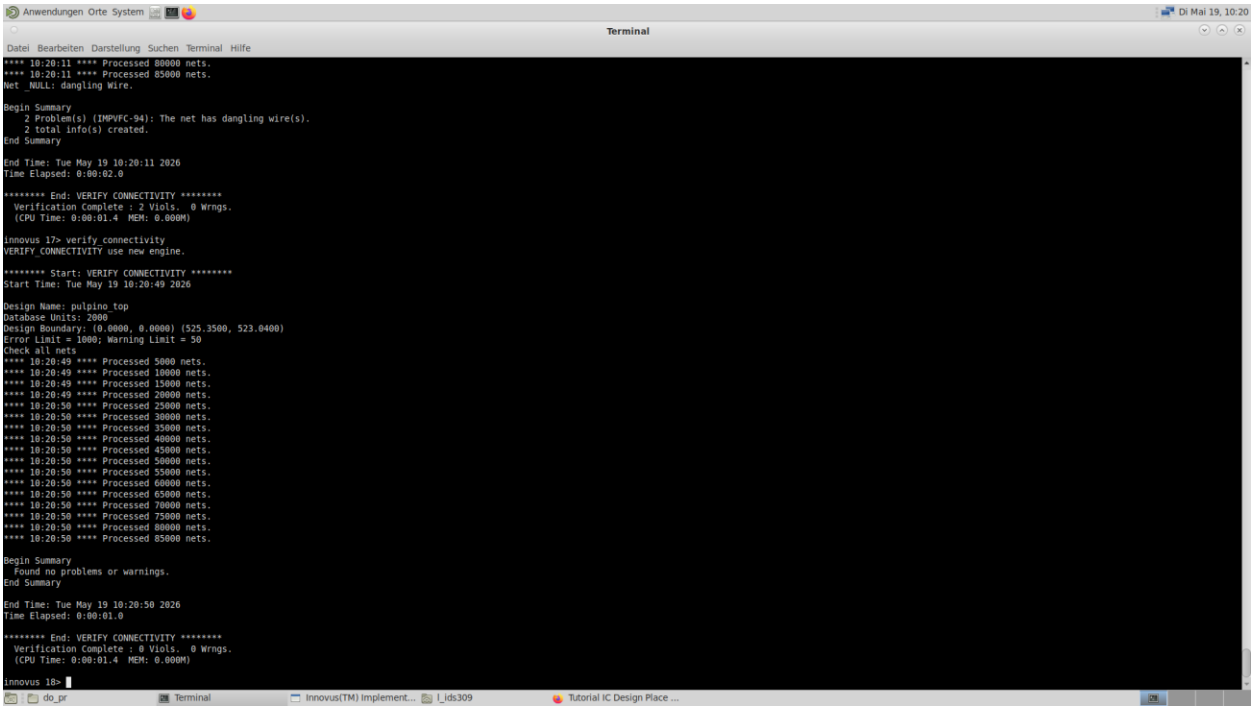
2.6 Verify and write result

After completing the routing steps, we did the design verification part where we checked is there any DRC and connectivity errors are present in the design. Design rule check is the process of checking physical layout data against fabrication-specific rules specified by the foundry to ensure successful fabrication.

DRC rules examples




Anwendungen Orte System
Di Mai 19, 10:19
Terminal
Datei Bearbeiten Darstellung Suchen Terminal Hilfe
VERIFY DRC ..... Sub-Area : 24 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {226.800 226.800 302.400 302.400} 25 of 49
VERIFY DRC ..... Sub-Area : 25 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {302.400 226.800 378.000 302.400} 26 of 49
VERIFY DRC ..... Sub-Area : 26 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {378.000 226.800 453.600 302.400} 27 of 49
VERIFY DRC ..... Sub-Area : 27 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {453.600 226.800 525.350 302.400} 28 of 49
VERIFY DRC ..... Sub-Area : 28 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {0.000 302.400 75.600 378.000} 29 of 49
VERIFY DRC ..... Sub-Area : 29 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {75.600 302.400 151.200 378.000} 30 of 49
VERIFY DRC ..... Sub-Area : 30 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {151.200 302.400 226.800 378.000} 31 of 49
VERIFY DRC ..... Sub-Area : 31 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {226.800 302.400 302.400 378.000} 32 of 49
VERIFY DRC ..... Sub-Area : 32 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {302.400 302.400 378.000 378.000} 33 of 49
VERIFY DRC ..... Sub-Area : 33 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {378.000 302.400 453.600 378.000} 34 of 49
VERIFY DRC ..... Sub-Area : 34 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {453.600 302.400 525.350 378.000} 35 of 49
VERIFY DRC ..... Sub-Area : 35 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {0.000 378.000 75.600 453.600} 36 of 49
VERIFY DRC ..... Sub-Area : 36 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {75.600 378.000 151.200 453.600} 37 of 49
VERIFY DRC ..... Sub-Area : 37 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {151.200 378.000 226.800 453.600} 38 of 49
VERIFY DRC ..... Sub-Area : 38 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {226.800 378.000 302.400 453.600} 39 of 49
VERIFY DRC ..... Sub-Area : 39 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {302.400 378.000 378.000 453.600} 40 of 49
VERIFY DRC ..... Sub-Area : 40 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {378.000 378.000 453.600 453.600} 41 of 49
VERIFY DRC ..... Sub-Area : 41 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {453.600 378.000 525.350 453.600} 42 of 49
VERIFY DRC ..... Sub-Area : 42 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {0.000 453.600 75.600 523.040} 43 of 49
VERIFY DRC ..... Sub-Area : 43 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {75.600 453.600 151.200 523.040} 44 of 49
VERIFY DRC ..... Sub-Area : 44 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {151.200 453.600 226.800 523.040} 45 of 49
VERIFY DRC ..... Sub-Area : 45 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {226.800 453.600 302.400 523.040} 46 of 49
VERIFY DRC ..... Sub-Area : 46 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {302.400 453.600 378.000 523.040} 47 of 49
VERIFY DRC ..... Sub-Area : 47 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {378.000 453.600 453.600 523.040} 48 of 49
VERIFY DRC ..... Sub-Area : 48 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {453.600 453.600 525.350 523.040} 49 of 49
VERIFY DRC ..... Sub-Area : 49 complete 0 Viols.
Verification Complete : 0 Viols.
*** End Verify DRC (CPU: 0:00:06.5 ELAPSED TIME: 7.00 MEM: 0.0M) ***
innovus 16>
do_pr
Terminal
Innovus(TM) Implement...
l_ids309
Tutorial IC Design Place ...


Anwendungen Orte System
Di Mai 19, 10:20
Terminal
Datei Bearbeiten Darstellung Suchen Terminal Hilfe
**** 10:20:11 **** Processed 80000 nets.
**** 10:20:11 **** Processed 85000 nets.
Net _NULL: dangling Wire.
Begin Summary
2 Problem(s) (IMPVFC-94): The net has dangling wire(s).
2 total info(s) created.
End Summary
End Time: Tue May 19 10:20:11 2026
Time Elapsed: 0:00:02.0
******** End: VERIFY CONNECTIVITY ********
Verification Complete : 2 Viols. 0 Wrngs.
(CPU Time: 0:00:01.4 MEM: 0.000M)
innovus 17> verify_connectivity
VERIFY_CONNECTIVITY use new engine.
******** Start: VERIFY CONNECTIVITY ********
Start Time: Tue May 19 10:20:49 2026
Design Name: pulpino_top
Database Units: 2000
Design Boundary: (0.0000, 0.0000) (525.3500, 523.0400)
Error Limit = 1000; Warning Limit = 50
Check all nets
**** 10:20:49 **** Processed 5000 nets.
**** 10:20:49 **** Processed 10000 nets.
**** 10:20:49 **** Processed 15000 nets.
**** 10:20:49 **** Processed 20000 nets.
**** 10:20:50 **** Processed 25000 nets.
**** 10:20:50 **** Processed 30000 nets.
**** 10:20:50 **** Processed 35000 nets.
**** 10:20:50 **** Processed 40000 nets.
**** 10:20:50 **** Processed 45000 nets.
**** 10:20:50 **** Processed 50000 nets.
**** 10:20:50 **** Processed 55000 nets.
**** 10:20:50 **** Processed 60000 nets.
**** 10:20:50 **** Processed 65000 nets.
**** 10:20:50 **** Processed 70000 nets.
**** 10:20:50 **** Processed 75000 nets.
**** 10:20:50 **** Processed 80000 nets.
**** 10:20:50 **** Processed 85000 nets.
Begin Summary
Found no problems or warnings.
End Summary
End Time: Tue May 19 10:20:50 2026
Time Elapsed: 0:00:01.0
******** End: VERIFY CONNECTIVITY ********
Verification Complete : 0 Viols. 0 Wrngs.
(CPU Time: 0:00:01.4 MEM: 0.000M)
innovus 18>
do_pr
Terminal
Innovus(TM) Implement...
l_ids309
Tutorial IC Design Place ...